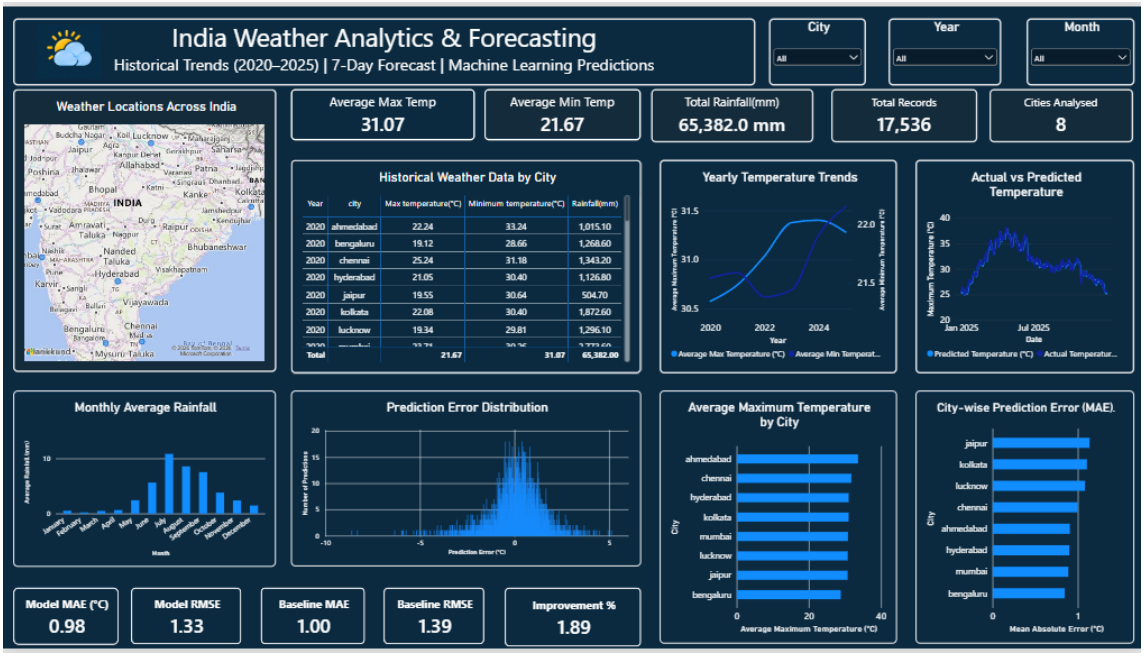
INDIA WEATHER ANALYTICS & FORECASTING

Project Report | Historical trends • 7-day forecast • Machine learning

Project by
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Tools
Python • PostgreSQL • Power BI • Scikit-learn • Open-Meteo API

Dashboard overview



This report explains the project scope, data, analysis, dashboard, model evaluation, and areas for improvement.

01 Executive Summary

This project explores weather patterns across eight Indian cities using daily historical observations from 2020 to 2025. I used Python to analyze the data, PostgreSQL to store and query it, and Power BI to present the findings. I also included a separate seven-day forecast from Open-Meteo and tested a machine-learning model for daily maximum-temperature prediction.

17,536	8	2020–2025
historical records	Indian cities	historical period

Project goals

- Compare temperature, rainfall, and wind patterns across cities.
- Explore yearly temperature trends and monthly rainfall patterns.
- Use SQL to summarize weather records and compare cities.
- Build a dashboard that makes the findings easier to explore.
- Evaluate a Random Forest model against a simple baseline.

Main takeaway

The project brings several types of analysis into one dashboard. The machine-learning model performed slightly better than the baseline, so the result is useful but the improvement is modest.

What is included

Component	Purpose
Historical analysis	City-level comparisons and trends from 2020–2025
API forecast	Separate seven-day weather forecast
Machine learning	One-day-ahead maximum-temperature prediction experiment
Power BI	Interactive summary of weather and model results

02 Data and Scope

The historical dataset contains daily observations for eight cities. The analysis uses date, city, temperature, wind-speed, precipitation, rain, and weather-code fields. The historical records and forecast records are treated as separate datasets.

Cities covered

City	City	City	City
Ahmedabad	Bengaluru	Chennai	Hyderabad
Jaipur	Kolkata	Lucknow	Mumbai

Data sources

Dataset	How it is used
Historical weather	Trend analysis, city summaries, rainfall and temperature comparisons
Open-Meteo forecast	Seven-day forecast values for the selected cities
ML evaluation data	Actual temperatures, model predictions, baseline predictions, and errors

Data checks that matter

- Keep the historical dataset within 2020–2025; do not mix 2026 forecast dates into historical summaries.
- Check that date columns are stored as dates and city names are consistent.
- Use Average for temperature when summarizing multiple records; do not add temperatures together.
- Use rainfall Sum only when the visual is intended to show accumulated rainfall.

The Open-Meteo forecast and the machine-learning predictions are different outputs. The API supplies the seven-day forecast; the model is evaluated for one-day-ahead predictions.

03 Tools and Workflow

Tool	Role in the project
Python	Data preparation, exploratory analysis, feature engineering, and evaluation
Pandas / NumPy	Data manipulation and calculations
Scikit-learn	Random Forest regression and evaluation metrics
PostgreSQL	Store data and run SQL summaries
Power BI	Dashboard, KPI cards, map, trends, and model visuals
Google Colab	Notebook environment for Python
Open-Meteo API	Retrieve separate seven-day forecasts

Project workflow

Stage	Work
1. Collect	Load historical weather data and retrieve forecast values.
2. Prepare	Check date types, city names, missing values, and data consistency.
3. Analyze	Compare city averages, rainfall totals, and trends.
4. Query	Use PostgreSQL for grouped summaries and SQL analysis.
5. Model	Train a Random Forest model and compare it with a baseline.
6. Present	Build the Power BI dashboard and document the findings.

The workflow connects data preparation, analysis, storage, modeling, and communication in one project.

04 Historical Weather Findings

The city summaries show differences in average maximum temperature. Rainfall totals show how much precipitation accumulated in the historical records over the selected period.

Average maximum temperature by city

City	Average maximum temperature
Ahmedabad	33.6°C
Chennai	31.8°C
Hyderabad	31.0°C
Kolkata	31.0°C
Mumbai	30.8°C
Lucknow	30.8°C
Jaipur	30.7°C
Bengaluru	28.8°C

Cities with the highest recorded rainfall totals

City	Total rainfall across historical period
Mumbai	14,945.7 mm
Kolkata	11,440.2 mm
Chennai	9,447.5 mm

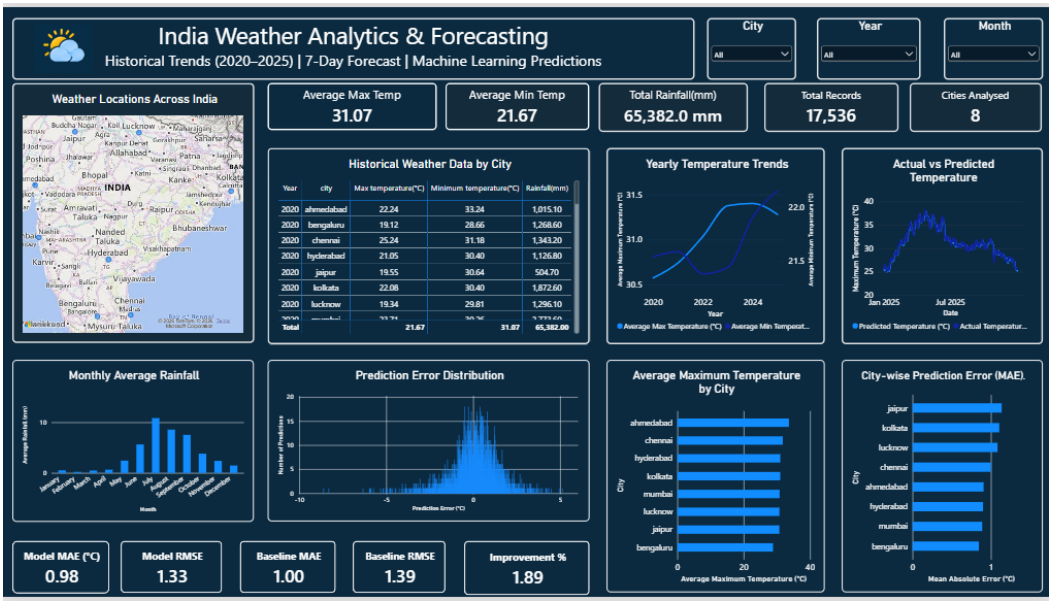
Interpretation

Ahmedabad has the highest average maximum temperature in the city summary, while Bengaluru has the lowest. Mumbai has the largest accumulated rainfall total among the eight cities in the historical dataset. These are descriptive comparisons; they do not explain the causes of the differences.

Temperature averages and period rainfall totals describe different things. Always label the aggregation and time period clearly when presenting these numbers.

05 Power BI Dashboard

The dashboard combines headline KPIs with visuals for geography, historical trends, and model evaluation. City, year, and month filters help users focus on the subset of data they want to inspect.



Dashboard components

Visual	What it shows
KPI cards	Average max/min temperature, rainfall, record count, and number of cities
India map	Locations of the eight cities
Historical table	Weather values grouped by city and date/year
Yearly temperature trends	Average maximum and minimum temperatures by year
Monthly average rainfall	Rainfall patterns by month
Actual vs predicted	Model predictions compared with observed temperatures
Error visuals	Prediction-error distribution and city-wise MAE

Before publishing, verify that the historical table only includes 2020–2025 records and that temperature fields use Average rather than Sum when summarizing multiple days.

06 Machine Learning Evaluation

I used a Random Forest regression model to predict daily maximum temperature. The input features included calendar information, previous-day temperature observations, previous rainfall, a recent temperature average, and city information.

Training and testing setup

Item	Setup
Training period	2020–2024
Testing period	2025
Target	Daily maximum temperature
Baseline	Previous day's maximum temperature

Evaluation results

Metric	Random Forest	Baseline
MAE	0.98°C	1.00°C
RMSE	1.33°C	1.39°C

The model's MAE is about 1.9% lower than the baseline. This is a small improvement, not a major performance gain.

How to interpret the metrics

- MAE is the average absolute difference between predicted and actual temperatures.
- RMSE penalizes larger errors more strongly than MAE.
- The test results are for the 2025 period and may not generalize to every year or unusual weather event.

Limitation

The evaluation is a one-day-ahead prediction experiment that assumes previous weather observations are available. It should not be described as the model generating the separate seven-day Open-Meteo forecast.

07 Conclusion and Next Steps

This project gave me practical experience working through a data analytics workflow: exploring weather records, writing SQL queries, testing a regression model, and presenting results in Power BI. The dashboard brings the key comparisons and model evaluation into one place.

Key takeaways

- The historical analysis covers 17,536 records across eight Indian cities from 2020 to 2025.
- City comparisons show differences in average maximum temperature and accumulated rainfall.
- The seven-day forecast is retrieved from Open-Meteo, separately from the machine-learning experiment.
- The Random Forest model slightly outperformed the baseline, with an MAE of 0.98°C versus 1.00°C.

Possible next steps

- Automate forecast and historical data refreshes.
- Include more cities and compare performance across seasons.
- Test additional models and evaluate across more test periods.
- Add the notebook, SQL scripts, dashboard screenshot, and instructions to GitHub.

Glossary

Term	Meaning
MAE	Mean Absolute Error; average size of prediction errors.
RMSE	Root Mean Squared Error; gives greater weight to larger errors.
Baseline	A simple reference method used to judge whether a model adds value.
API forecast	Forecast data retrieved from an external weather service.

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Project tools: Python, PostgreSQL, Power BI, and machine learning.